

1. Write a program to generate and visualize the following distributions. **(Do not use the predefined library functions.)**

- Normal Distributions

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Where σ is variance and μ is the mean.

- Uniform Distribution

$$f(x) = \begin{pmatrix} \frac{1}{b-a} & \text{if } a < x < b; \\ 0 & \text{otherwise} \end{pmatrix}$$

Where a and b are the boundaries.

- Exponential Distribution

$$f(x) = \begin{pmatrix} \lambda e^{-\lambda x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{pmatrix}$$

Where λ is the rate parameter.

- Poisson Distribution

$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

Where λ is the rate parameter.

- Cauchy Distribution

$$f(x) = \frac{1}{\pi(1+x^2)}$$

- t-Distribution

$$t = \frac{\bar{x} - \mu}{s\sqrt{n}}$$

Where $\bar{x}$ is sample mean, μ is population mean, s is sample standard deviation and n is sample size.

- Chi-squared Distribution

$$f(x) = \begin{cases} \frac{x^{\frac{k}{2}-1} e^{-\frac{x}{2}}}{2^{\frac{k}{2}} \Gamma(\frac{k}{2})} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Where k is degrees of freedom and $\Gamma(\frac{k}{2})$ is gamma function.

2. Considering iris dataset ([Click here to download](#)), D of size $N \times M$, with N: number of samples and M: number of features, design a Bayesian classifier using posterior probabilities. Divide the dataset into training and testing data in (a) 70:30 ratio and (b) 80:20 ratio. Calculate the accuracy and error rate. **(Do not use predefined library functions.)**